The sexually transmitted disease model for cervical cancer: incoherent epidemiologic findings and the role of misclassification of human papillomav...

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The definition of cervical neoplasia as a sexually determined disease caused by some types of human papillomavirus has been widely accepted. Recent epidemiologic studies, however, have failed to identify a correlation between sexual activity and human papillomavirus infection. Moreover, sexual activity has also been shown to be independent of human papillomavirus infection in increasing cervical cancer risk. These incoherences are analyzed with respect to etiologic models for cervical neoplasia and by considering the role of misclassification of human papillomavirus infection in interpreting the relations assumed under these models. Even small levels of misclassification can considerably distort (1) the presumed prevalence of viral infection, (2) the association between sexual activity and human papillomavirus infection, and (3) the ability to control the relation between sexual activity and cancer by human papillomavirus infection. In field surveys, the presumed rates of human papillomavirus infection based on a DNA assay such as the filter in situ hybridization may be a gross overestimation of the true viral prevalence. Use of moderately misclassified human papillomavirus infection test results for effect estimation and covariate adjustment in data analysis may seriously distort the underlying relations. Consequently, considering these conditions, the apparent incoherence of recent epidemiologic findings should not be construed as evidence against the role of human papillomavirus in the etiology of cervical cancer or the validity of the sexually transmitted disease model.